

Multi-Band Spectrum Analysis & Parametric Trajectories

Mathematical Models of [spectrometer~] and [time.xy] Visualizers

Time Dilation DAW (v0.0.1) Engineering Group

August 3, 2026

Abstract

This document formulates the 32-band logarithmic frequency estimation, peak ballistics decay dynamics, and 2D parametric state phase space trajectory plotting powering the [spectrometer~] and [time.xy] visualizer node objects in **Time Dilation DAW**.

1 Logarithmic Frequency Band Spacing

Human auditory perception of pitch follows logarithmic frequency scaling (octaves). In the 32-band live spectrum estimator [spectrometer~], band center frequencies f_b for $b \in [0, 31]$ are spaced logarithmically between $f_{\min} = 20$ Hz and $f_{\max} = 20\,000$ Hz:

$$f_b = f_{\min} \cdot \left(\frac{f_{\max}}{f_{\min}} \right)^{\frac{b}{B-1}}, \quad \text{where } B = 32 \quad (1)$$

2 Short-Time Energy & Peak Ballistics Decay

For each block of audio input $x(n)$ of length K :

$$E_{\text{raw}} = \max_{n=0}^{K-1} |x(n)| \quad (2)$$

For band b , band magnitude level $M_b(k)$ at block k is smoothed using 1-pole lowpass filter:

$$M_b(k) = 0.75 \cdot M_b(k-1) + 0.25 \cdot T_b(k) \quad (3)$$

where target $T_b(k) = E_{\text{raw}} \cdot \left(1 - 0.4 \frac{b}{B-1} \right) \cdot \Phi_b(t)$.

2.1 Peak Indicator Hold & Exponential Decay

Peak indicator dot levels $P_b(k)$ catch instant maxima and decay exponentially:

$$P_b(k) = \begin{cases} M_b(k), & M_b(k) \geq P_b(k-1) \\ 0.94 \cdot P_b(k-1), & M_b(k) < P_b(k-1) \end{cases} \quad (4)$$

3 2D Phase Space Parametric Trajectories ([time.xy])

The [time.xy] oscilloscope node visualizes the 2D parametric state space trajectory $\mathbf{r}(t) = (x(t), y(t))^T$ generated by two audio or control signal streams:

$$\mathbf{r}[n] = \begin{bmatrix} x[n] \\ y[n] \end{bmatrix} \in [-2.0, 2.0] \times [-2.0, 2.0] \quad (5)$$

Points are stored in a 256-sample ring buffer:

$$\mathcal{P} = \{\mathbf{r}[n - 255], \dots, \mathbf{r}[n - 1], \mathbf{r}[n]\} \quad (6)$$

Rendering a continuous vector trace path on the dark slate canvas matrix allows real-time visualization of Lissajous curves, phase relationships, and relativistic coordinate time orbits.